



VISUAL CODE OPERATIONS

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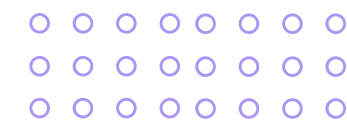


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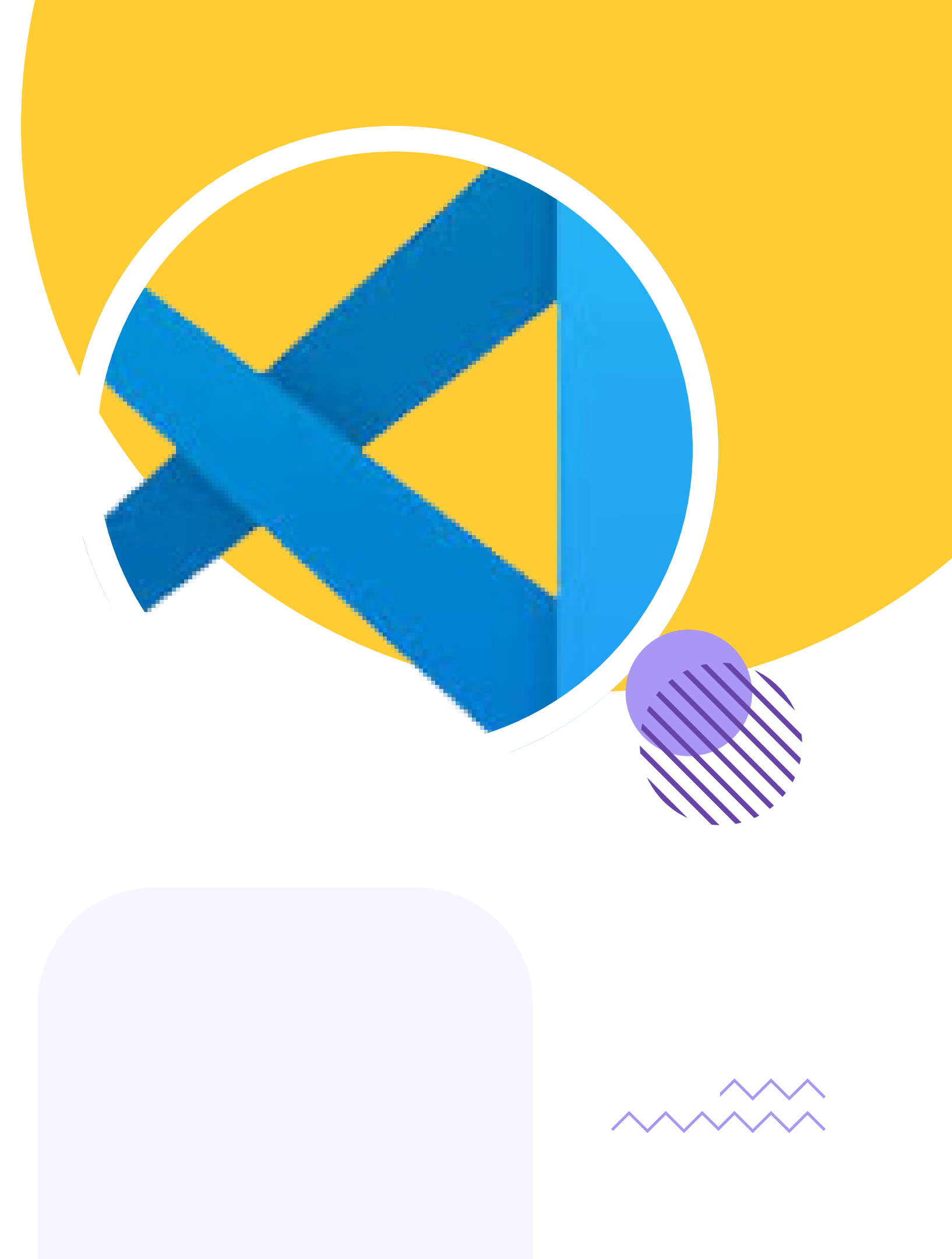
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DEFINITION

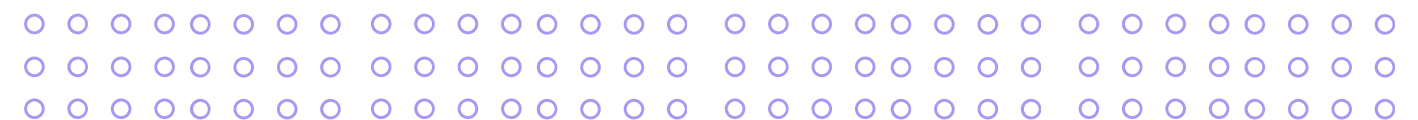
Visual Studio Code (VS Code) merupakan salah satu editor kode modern yang banyak digunakan oleh pengembang aplikasi karena kemudahan, fleksibilitas, dan fitur-fiturnya yang lengkap. VS Code dirancang sebagai editor ringan namun memiliki kemampuan yang mendekati Integrated Development Environment (IDE).

```
filePath = filePath + "\\  
# Check for confirmation  
questionDialogText = "Obj Sequence will be saved as:\n\n\  
" + filePath + objName + "###.obj\n\n\  
"From frame " + str(fromTime) + " to " + str(toTime) + " for " + str(animLength) + " frames."  
proceedBool = c4d.gui.QuestionDialog(questionDialogText)  
  
if proceedBool == True:  
  
    # loop through animation and export frames  
    for x in range(0, animLength):  
  
        # change frame, redraw view  
        moveTime = c4d.BaseTime(fromTime, docFps) + c4d.BaseTime(x, docFps)  
        doc.SetTime(moveTime)  
        c4d.EventAdd(c4d.EVENT_FORCEREDRAW)  
        c4d.DrawViews(c4d.DRAWFLAGS_FORCEFULLREDRAW)
```



WHAT IS GUI TKINTER

GUI Tkinter adalah library bawaan Python yang digunakan untuk membuat antarmuka grafis (Graphical User Interface). Dengan Tkinter, dapat membuat aplikasi dengan elemen visual seperti tombol, kotak teks, label, jendela input, hingga layout grid. Tkinter bekerja dengan konsep widget, yaitu komponen-komponen UI yang dapat ditambahkan dan dikonfigurasi sesuai kebutuhan. Setiap widget bisa diatur warna, ukuran, posisi, hingga fungsinya ketika ditekan atau diisi.



THE REASON CHOOSE CALCULATOR

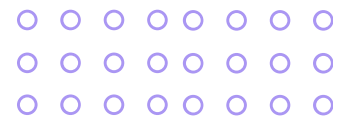
Saya memilih aplikasi kalkulator sebagai pemenuhan akhir mata kuliah. Karena, aplikasi ini mencakup konsep fundamental pemrograman yang penting, seperti pembuatan fungsi, pengolahan input-output, penggunaan percabangan, serta penerapan antarmuka grafis (GUI) menggunakan Tkinter.





GitHub

[HTTPS://GITHUB.COM/GIANN13Z/PROGRAMAPK.GIT](https://github.com/GIANN13Z/PROGRAMAPK.GIT)





01 IMPORT LIBRARY

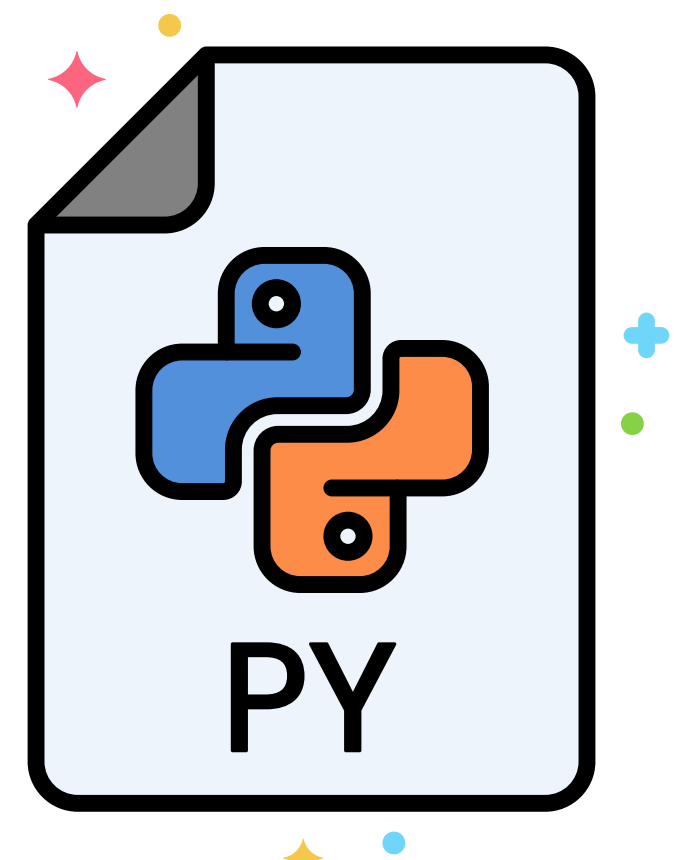
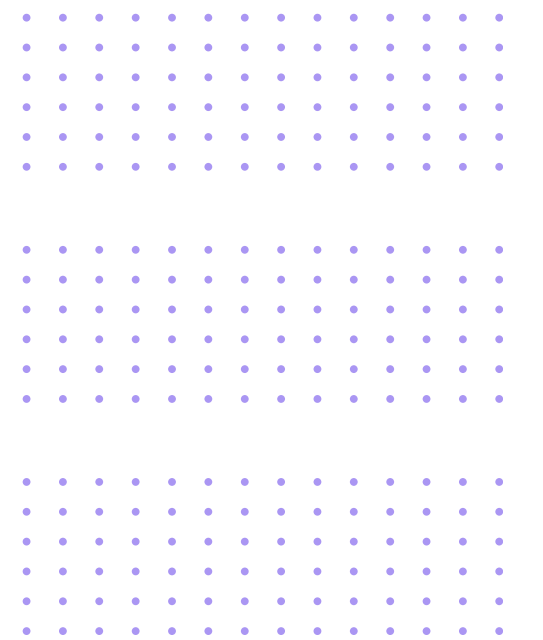
```
1  from tkinter import*
2  import tkinter.font as font
3  import math
```

02 MAIN WINDOW

```
root = Tk()
root.title("Kalkulator")
root["bg"] = "#d1d1d1"
root.geometry("310x400")
```

03 FONT

```
myfont = font.Font(size=15)
```



04

WIDGET ENTERY

```
e = Entry(root,width=25,borderwidth=0)
e["font"]= myfont
e["bg"] =  "#d1d1d1"
e.grid(row = 0,columnspan=4,pady=20,padx=20)
```

05

INPUT ANGKA

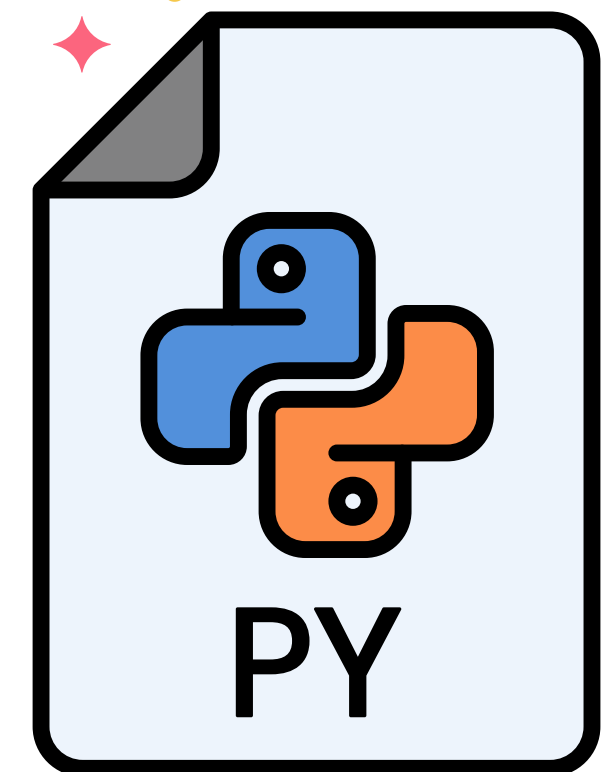
```
17 def cetak(nilai):
18     current = e.get()
19     e.delete(0,END)
20     e.insert(0, str(current)+str(nilai))
21
```

06

ARITHMETIC OPERATIONS

```
22 def tambah():
23     nomor_awal = e.get()
24     global n_awal
25     global mtk
26     mtk = "penjumlahan"
27     n_awal = int(nomor_awal)
28     e.delete(0,END)
```

SYNTAX



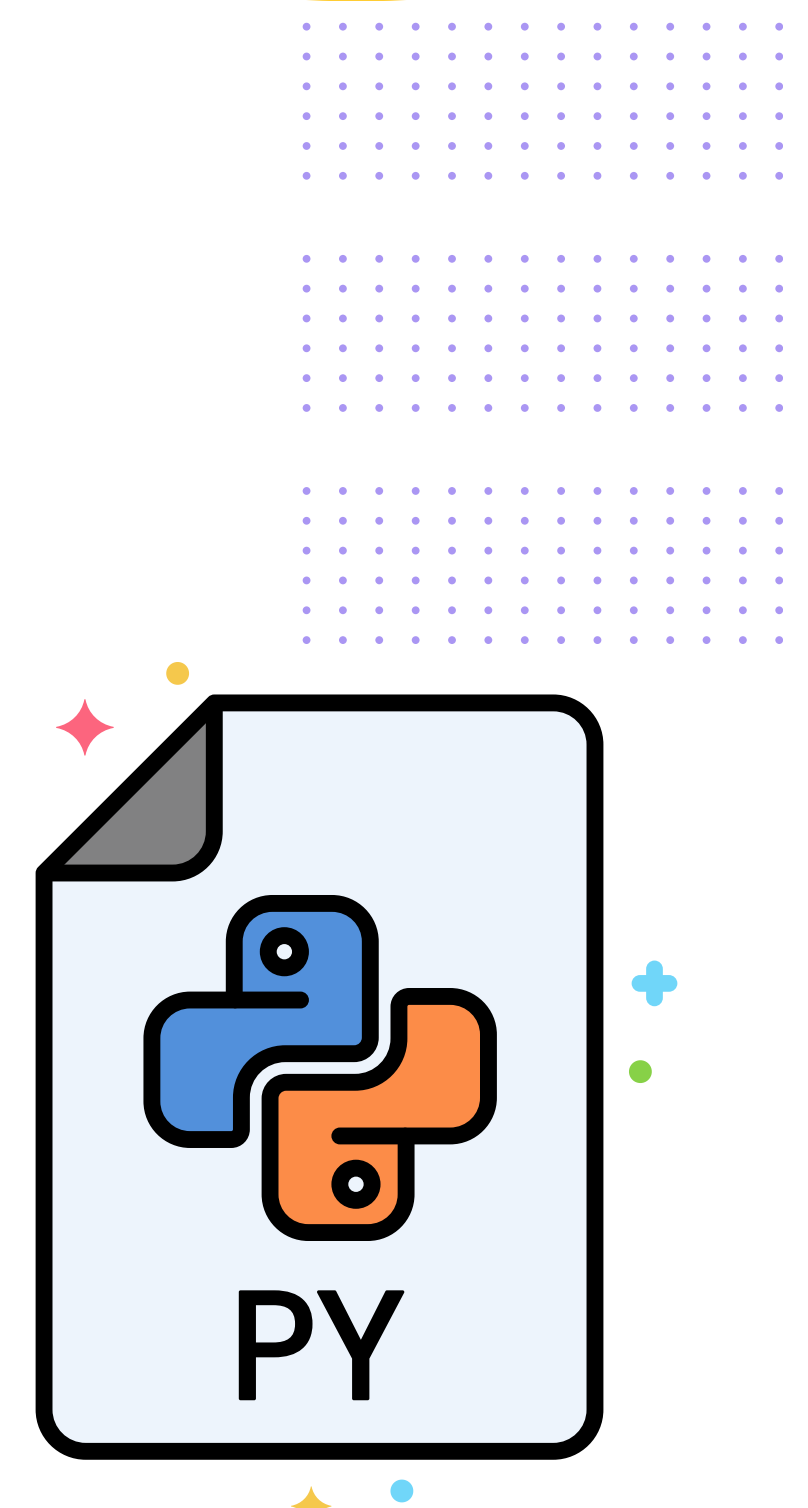


07 ARITHMETIC OPERATIONS

```
74 def hapus():
75     e.delete(0,END)
```

08 ARITHMETIC OPERATIONS

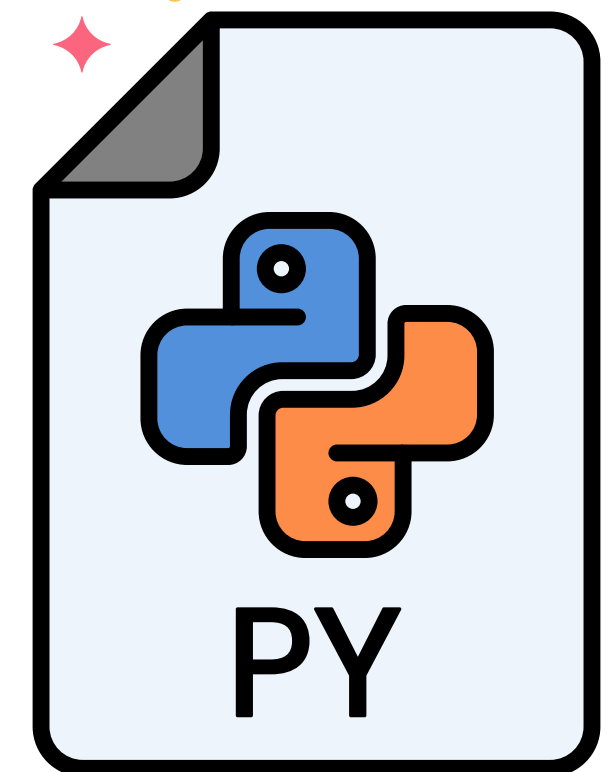
```
76 def samadengan():
77     nomor_akhir = e.get()
78     e.delete(0,END)
79     if mtk == "penjumlahan":
80         e.insert(0,n_awal + int(nomor_akhir))
81     elif mtk == "pengurangan":
82         e.insert(0,n_awal - int(nomor_akhir))
83     elif mtk == "pembagian":
84         try:
85             hitung = n_awal / int(nomor_akhir)
86             e.insert(0,hitung)
87         except ZeroDivisionError:
88             e.insert(0,"maaf komputer anda kentang")
89
90     elif mtk == "perkalian":
91         e.insert(0,n_awal * int(nomor_akhir))
92     elif mtk == "persen":
93         angka_akhir = float(nomor_akhir)
94         hasil = n_awal * (angka_akhir / 100)
95         e.insert(0, hasil)
96
```



SYNTAX

11 BUTTON FUNCTIONS

```
97 btn1 = Button(root,text="1",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(1))
98 btn2 = Button(root,text="2",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(2))
99 btn3 = Button(root,text="3",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(3))
100 btn4 = Button(root,text="4",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(4))
101 btn5 = Button(root,text="5",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(5))
102 btn6 = Button(root,text="6",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(6))
103 btn7 = Button(root,text="7",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(7))
104 btn8 = Button(root,text="8",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(8))
105 btn9 = Button(root,text="9",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(9))
106 btn0 = Button(root,text="0",padx = 30,bg="#3d3d3d",fg="white", pady = 20,command=lambda:cetak(0))
107 tam = Button(root,text="+",padx = 30,bg="#FFA500",fg="white", pady = 20,command=tambah)
108 kur = Button(root,text="-",padx = 30,bg="#FFA500",fg="white", pady = 20,command=kurang)
109 bag = Button(root,text="/",padx = 30,bg="#FFA500",fg="white", pady = 20,command=bagi)
110 kal = Button(root,text="x",padx = 30,bg="#FFA500",fg="white", pady = 20,command=kali)
111 pang = Button(root,text="x2",padx = 30,bg="#878787",fg="white", pady = 20,command=pangkat)
112 ak2 = Button(root,text="sq2",padx = 25,bg="#878787",fg="white", pady = 20,command=akar)
113 pers = Button(root,text="%",padx = 30,bg="#878787",fg="white", pady = 20,command=persen)
114 hap = Button(root,text="C",padx = 30,bg="#878787",fg="white", pady = 20,command=hapus)
115 equal = Button(root,text="=",padx = 60,bg="#FFA500", pady = 20,command=samadengan)
```



12

BUTTON GRID

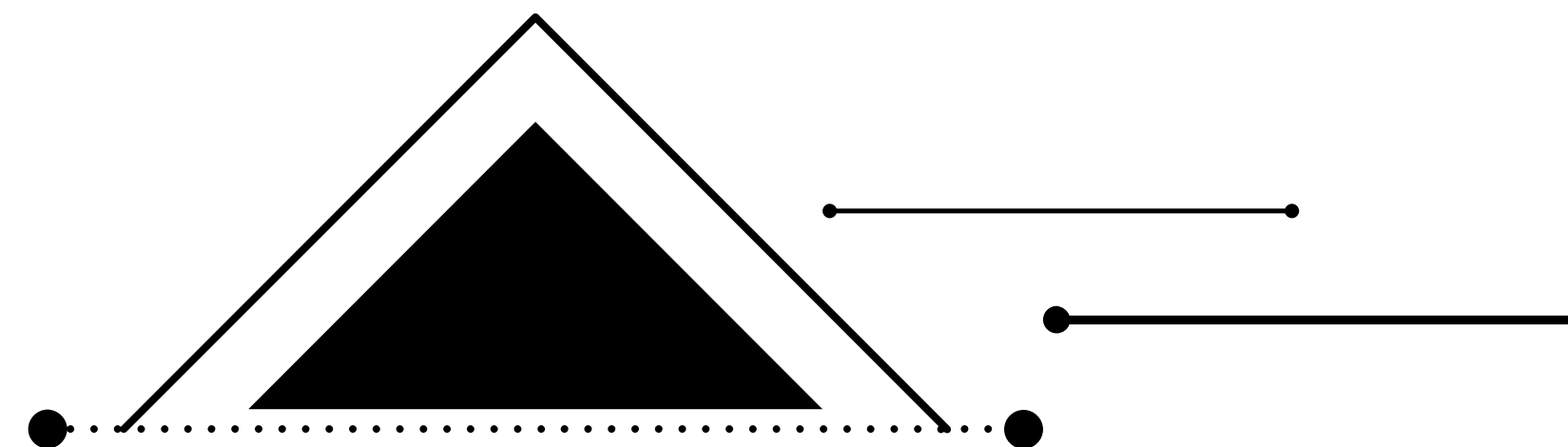
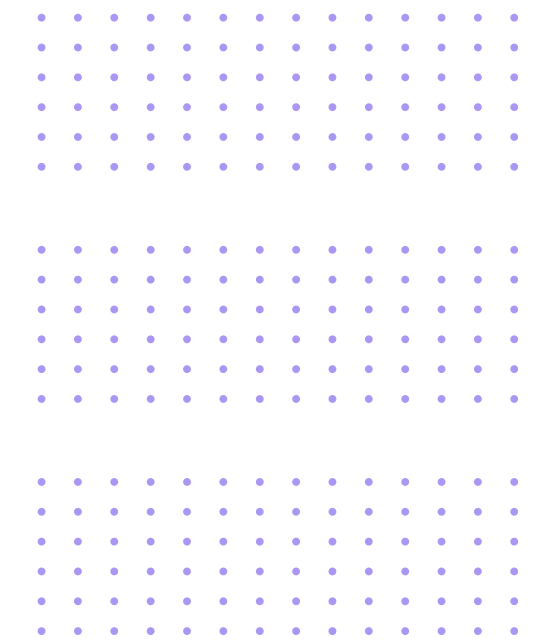
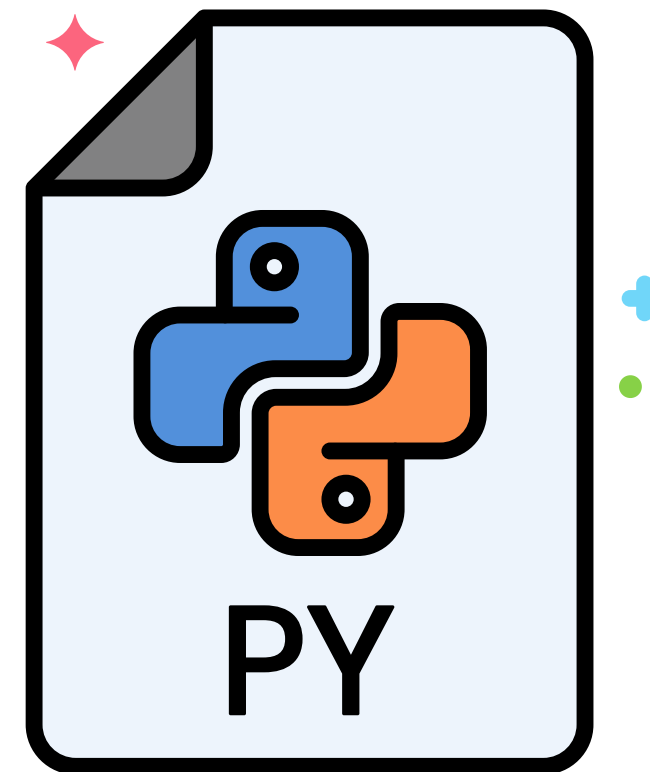
```
117 btn1.grid(row=3,column=0,pady=2)
118 btn2.grid(row=3,column=1,pady=2)
119 btn3.grid(row=3,column=2,pady=2)
120 btn4.grid(row=2,column=0,pady=2)
121 btn5.grid(row=2,column=1,pady=2)
122 btn6.grid(row=2,column=2,pady=2)
123 btn7.grid(row=1,column=0,pady=2)
124 btn8.grid(row=1,column=1,pady=2)
125 btn9.grid(row=1,column=2,pady=2)
126 btn0.grid(row=4,column=1,pady=2)
127
128 tam.grid(row=1,column=3,pady=2)
129 kur.grid(row=2,column=3,pady=2)
130 bag.grid(row=3,column=3,pady=2)
131 kal.grid(row=4,column=3,pady=2)
132 hap.grid(row=4,column=0,pady=2)
133 equal.grid(row=5,column=2,columnspan=2)
134 pang.grid(row =5,column=0,pady=2)
135 ak2.grid(row =5,column=1,pady=2)
136 pers.grid(row =4,column=2,pady=2)
```

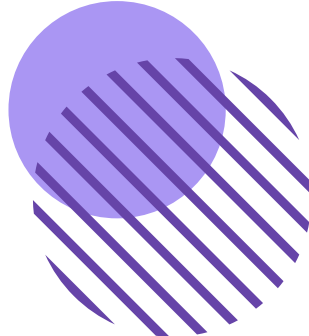
13

LOOPING GRAPH

```
137
138 root.mainloop()
139
```

SYNTAX





CONCLUSION

penggunaan pemrograman dalam pengembangan aplikasi memberikan kontribusi besar dalam menciptakan solusi digital yang praktis dan efisien. Visual Studio Code sebagai editor kode, Python sebagai bahasa pemrograman, serta Tkinter sebagai library antarmuka grafis merupakan kombinasi yang ideal bagi pemula maupun pengembang untuk membuat aplikasi seperti kalkulator maupun aplikasi form seperti buku tamu.

